OSPF (Part-2)

- Topics covered:
    - OSPF Metric
    - Becoming OSPF Neighbours
    - More OSPF Configuration

OSPF Metric (Cost)

What Is OSPF Metric?

OSPF's metric is called **cost**.

It is automatically calculated based on the bandwidth (speed) of the interface.

You can also manually configure the cost of an interface, which will be shown later.

How OSPF Cost Is Calculated:

The interface cost is calculated using this formula:

OSPF Cost = Reference Bandwidth / Interface Bandwidth

- Default reference bandwidth: 100 megabits per second

Examples:

- Ethernet (10 Mbps):
  100 ÷ 10 = 10
- Fast Ethernet (100 Mbps):
  100 ÷ 100 = 1
- Gigabit Ethernet (1000 Mbps):
  100 ÷ 1000 = 0.1 → converted to 1
- 10-Gig Ethernet (10000 Mbps):
  100 ÷ 10000 = 0.01 → converted to 1

In OSPF:

- Any value less than 1 becomes 1

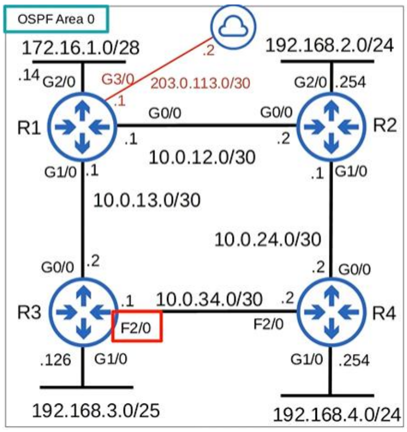
Therefore:

- Fast Ethernet
- Gigabit Ethernet
- 10-Gig Ethernet

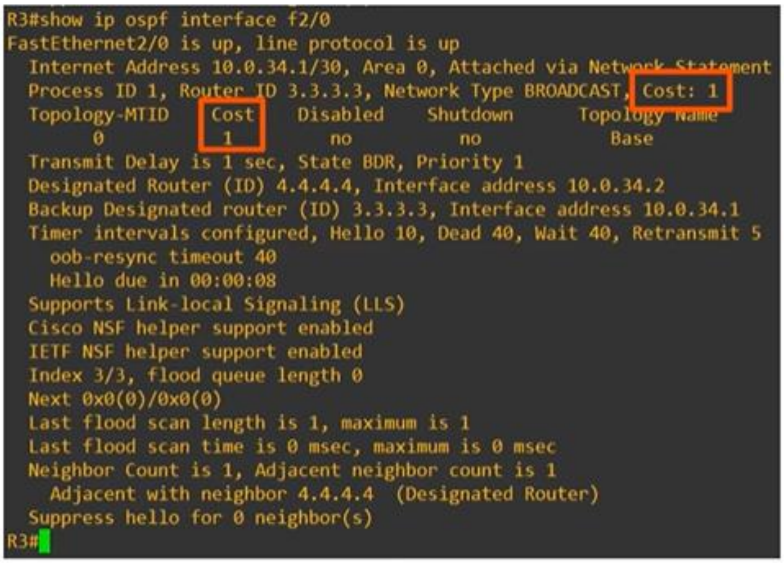
All have a cost of 1 by default.

Default Cost Shown in CLI (Fast Ethernet)

Network Topology (R3 F2/0 highlighted):



CLI:

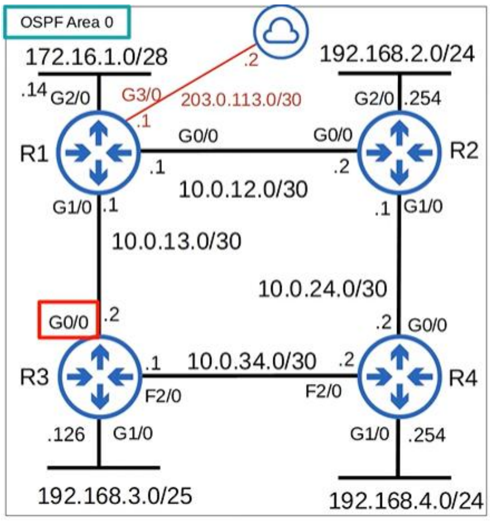


```
R3#show ip ospf interface f2/0
FastEthernet2/0 is up, line protocol is up
  Internet Address 10.0.34.1/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 4.4.4.4, Interface address 10.0.34.2
  Backup Designated router (ID) 3.3.3.3, Interface address 10.0.34.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:08
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 4.4.4.4  (Designated Router)
  Suppress hello for 0 neighbor(s)
R3#
```

The cost appears in two places, and the default Fast Ethernet cost is 1.

Default Cost Shown in CLI (Gigabit Ethernet)

Network Topology (R3 G0/0 highlighted):



CLI:

```
R3#show ip ospf interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.0.13.2/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 3.3.3.3, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 10.0.13.2
  Backup Designated router (ID) 1.1.1.1, Interface address 10.0.13.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 2
  Last flood scan time is 0 msec, maximum is 4 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1  (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
R3#
```

Both Fast Ethernet and Gigabit Ethernet have a cost of 1 by default.

This default situation is not ideal.

Changing the Reference Bandwidth

You can change the reference bandwidth from OSPF configuration mode.

CLI:

```
R3(config-router)#auto-cost reference-bandwidth ?
  <1-4294967>  The reference bandwidth in terms of Mbits per second

R3(config-router)#auto-cost reference-bandwidth 100000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
R3(config-router)#
```

- Reference bandwidth is configured in megabits per second
- 100,000 Mbps = 100 Gbps

New Cost Values After Changing Reference Bandwidth:

With reference bandwidth set to 100,000 Mbps:

- Fast Ethernet (100 Mbps):
  100,000 ÷ 100 = **1000**
- Gigabit Ethernet (1000 Mbps):
  100,000 ÷ 1000 = **100**

The reference bandwidth should be:

- Higher than the fastest link
- To allow for future upgrades

The same reference bandwidth must be configured on all OSPF routers.

OSPF Cost to a Destination

The OSPF cost to a destination is the total cost of outgoing interfaces.

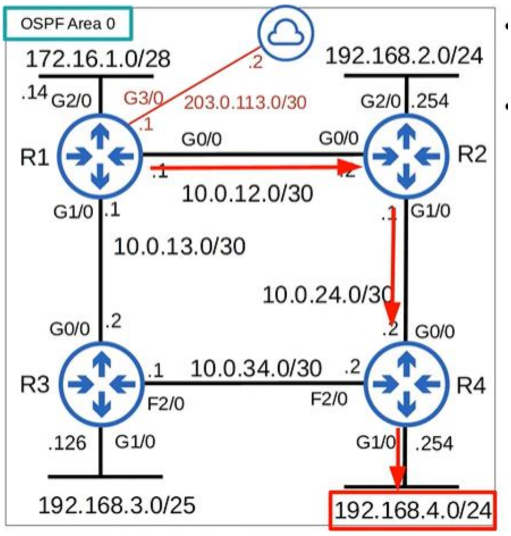
This is like Spanning Tree cost.

Example: Cost to 192.168.4.0/24

Path taken:

- R1 G0/0
- R2 G1/0
- R4 G1/0

Cost calculation: 100 + 100 + 100 = 300



Loopback Interface Cost

Loopback interfaces have a cost of 1.

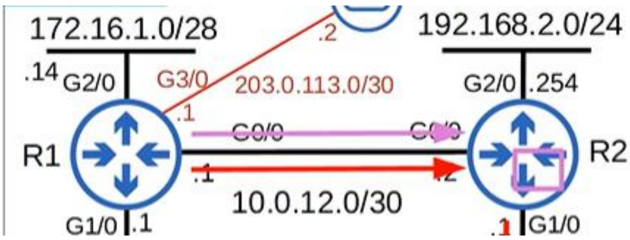
Example: Cost to 2.2.2.2 (R2 Loopback)

Path:

- R1 G0/0 (100)
- R2 Loopback (1)

Total cost:

100 + 1 = 101



Routing Table Before Changing Reference Bandwidth

CLI (Before):

O 192.168.4.0/24 [110/2] via 10.0.12.2

O 192.168.4.0/24 [110/2] via 10.0.13.2

```
O     192.168.4.0/24 [110/3] via 10.0.13.2, 00:00:04, GigabitEthernet1/0
                     [110/3] via 10.0.12.2, 00:00:04, GigabitEthernet0/0
```

- Two equal-cost routes
- Fast Ethernet and Gigabit Ethernet treated the same

Routing Table After Changing Reference Bandwidth CLI (After):

O 192.168.4.0/24 [110/300] via 10.0.12.2

```
O     192.168.4.0/24 [110/300] via 10.0.12.2, 00:26:46, GigabitEthernet0/0
```

- Only one route installed
- Total cost is 300

Cost to 2.2.2.2 is 101, as calculated earlier.

```
2.2.2.2 [110/101] via 10.0.12.2, 00:34:04, GigabitEthernet0/0
```

Manually Configuring OSPF Cost

You can manually configure the cost of an interface.

This cost overrides the auto-calculated cost.

CLI:

```
R1(config)#interface g0/0
R1(config-if)#ip ospf cost ?
  <1-65535>  Cost

R1(config-if)#ip ospf cost 10000
```

Verification:

```
R1(config-if)#do show ip ospf interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.0.12.1/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 10000
  Topology-MTID    Cost    Disabled    Shutdown    Topology Name
        0          10000      no          no          Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 10.0.12.2
  Backup Designated router (ID) 1.1.1.1, Interface address 10.0.12.1
```

Changing Interface Bandwidth

Another way to change OSPF cost is to change the interface bandwidth.

Formula reminder:

OSPF Cost = Reference Bandwidth / Interface Bandwidth

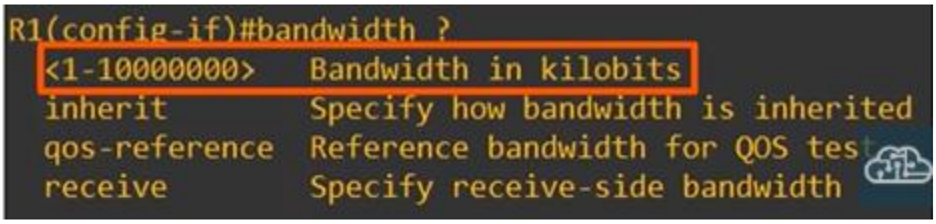
Important Clarification:

- Speed ≠ Bandwidth
- Changing bandwidth does not change actual interface speed
- Bandwidth is a logical value used for:
  - OSPF cost
  - EIGRP metric
  - Other calculations

To change actual speed, use the SPEED command.

BANDWIDTH Command:

CLI:



- Bandwidth command uses kilobits per second
- Reference bandwidth uses megabits per second

Always check units using ?.

Changing interface bandwidth is not recommended for changing OSPF cost.

Summary: Three Ways to Modify OSPF Cost

- Three ways to modify the OSPF cost:

  1) Change the **reference bandwidth**:
  R1(config-router)# **auto-cost reference-bandwidth** *megabits-per-second*

  2) Manual configuration
  R1(config-if)# **ip ospf cost** *cost*

  3) Change the **interface bandwidth**
  R1(config-if)# **bandwidth** *kilobits-per-second*

Checking OSPF Cost Quickly

```
R3#show ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo0          1     0               3.3.3.3/32         1     LOOP  0/0
Gi1/0        1     0               192.168.3.126/25   100   DR    0/0
Fa2/0        1     0               10.0.34.1/30       1000  BDR   1/1
Gi0/0        1     0               10.0.13.2/30       100   DR    1/1
```

This command provides a quick overview of OSPF-enabled interfaces and their costs.

Becoming OSPF Neighbours

Introduction to OSPF Neighbours

This is another very important topic in OSPF: OSPF neighbours.

Making sure routers successfully become OSPF neighbours is the main task in configuring and troubleshooting OSPF.

Once routers become neighbours, they automatically:

- Share network information
- Calculate routes

So, you only need to ensure:

- OSPF is activated on the correct interfaces
- Proper conditions are met to allow neighbour formation

More advanced OSPF configurations exist, but they are not necessary for basic OSPF operation.

If routers cannot become OSPF neighbours, OSPF cannot operate at all, so this topic is very important.

How Routers Become OSPF Neighbours

When OSPF is activated on an interface:

- The router sends OSPF Hello messages
- Hellos are sent at regular intervals (hello timer)

Hello messages are used to:

- Introduce the router to potential neighbours
- Check compatibility
- Negotiate the neighbour relationship

Default hello timer on Ethernet connections is 10 seconds.
Remember this number.

OSPF Hello messages are multicast to **224.0.0.5**, the multicast address for all OSPF routers.

Multicast comparison:

- RIP → 224.0.0.9
- EIGRP → 224.0.0.10
- OSPF → 224.0.0.5

OSPF messages are encapsulated in IP packets with:

- IP protocol number **89**

Overview of OSPF Neighbour States:

Routers must pass through several neighbour states.

This is a basic overview but contains a lot of information.

Assume:

- OSPF is already active on R2 G0/0
- OSPF is then activated on R1 G0/0

All State of OSPF Neighbours

Down State:

R1 sends an OSPF Hello message to 224.0.0.5.

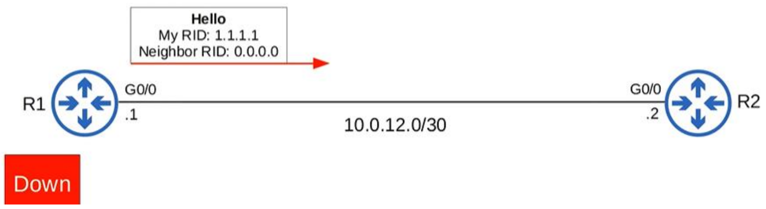
Important Hello fields:

- R1's router ID
- Neighbour router ID

R1 does not know about R2 yet, so:

- Neighbour router ID = 0.0.0.0

R1 has no neighbours yet, so the state is Down.



This is the first OSPF neighbour state, Down.

Init State:

When R2 receives the Hello packet:

- It adds R1 to its OSPF neighbour table

In R2's neighbour table:

- The relationship with R1 is now **Init**

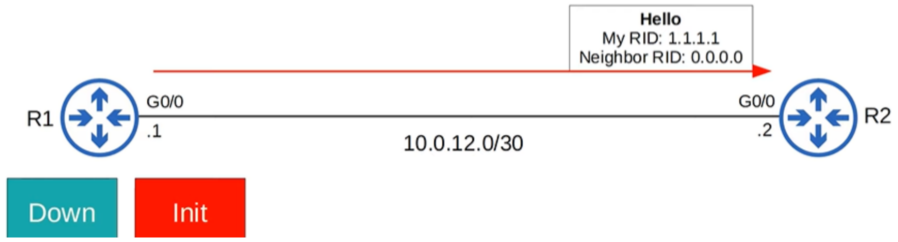
R1 still does not know about R2, so:

- R1 has no neighbour entries yet

Init state definition:

- A Hello packet was received
- The router's own RID is not in the Hello packet

R2's RID is 2.2.2.2
But R1's Hello contained neighbour RID 0.0.0.0



State: - R1: Down, R2: Init

2-Way State:

R2 sends a Hello packet containing:

- R1's RID
- R2's RID

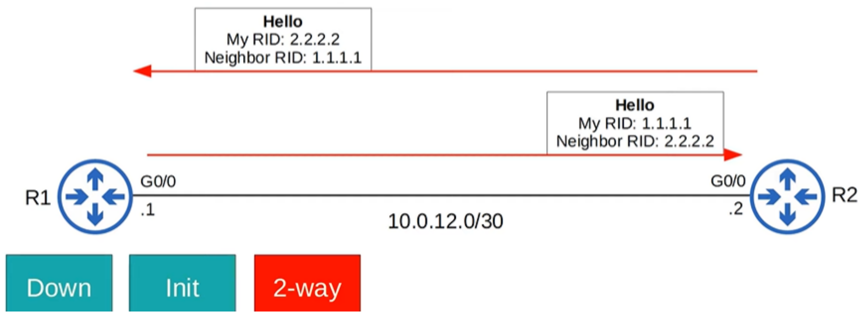
R1 receives it and:

- Inserts R2 into its neighbour table
- Sets state to 2-way

R1 then sends another Hello:

- This time including R2's RID

Now both routers are in the 2-way state.



State: - R1: 2-way R2: 2-way

2-way state meaning:

- The router received a Hello containing its own RID

If routers reach 2-way:

- All conditions to become neighbours are met
- They are ready to exchange LSAs

If they fail to reach 2-way:

- Troubleshooting is required

In some network types:

- DR and BDR elections occur here
  (This will be covered later.)

At this point:

- Routers are already OSPF neighbours
- Next states build a full adjacency

Exstart State:

After 2-way:

- Routers prepare to exchange LSDB information
- They must decide who starts the exchange

They choose:

- Master
- Slave

This is different from DR and BDR.

Decision happens in the Exstart state.

Rules:

- Higher RID → Master
- Lower RID → Slave

In this case:

- R2 (RID 2.2.2.2) → Master
- R1 (RID 1.1.1.1) → Slave

They exchange DBD (Database Description) packets to decide.

Result: - R2 = Master, R1 = Slave

Exstart is preparation for the next state.

Exchange State:

In the Exchange state:

- Routers exchange DBD packets
- DBDs contain a list of LSAs in the LSDB

DBDs:

- Do not contain full LSAs
- Only basic summary information

Routers compare:

- Received DBD
- Their own LSDB

They determine:

- Which LSAs they need from the neighbour

After exchanging DBDs, routers move to the next state.

Loading State:

In the Loading state:

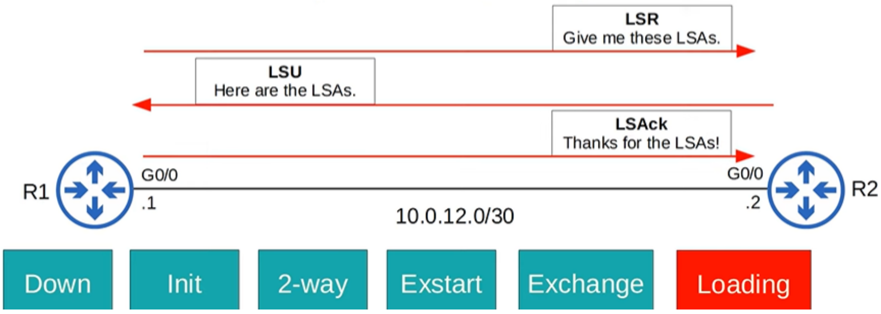
- Routers send LSR (Link State Request) messages
- They request LSAs they are missing

LSAs are sent in:

- LSU (Link State Update) messages

After receiving LSAs:

- Routers send LSAck messages to acknowledge receipt



After this:

- Both routers have identical LSDBs

Full State

This is the final OSPF neighbour state.

In the Full state:

- Routers have a full OSPF adjacency
- LSDBs are identical

They continue to:

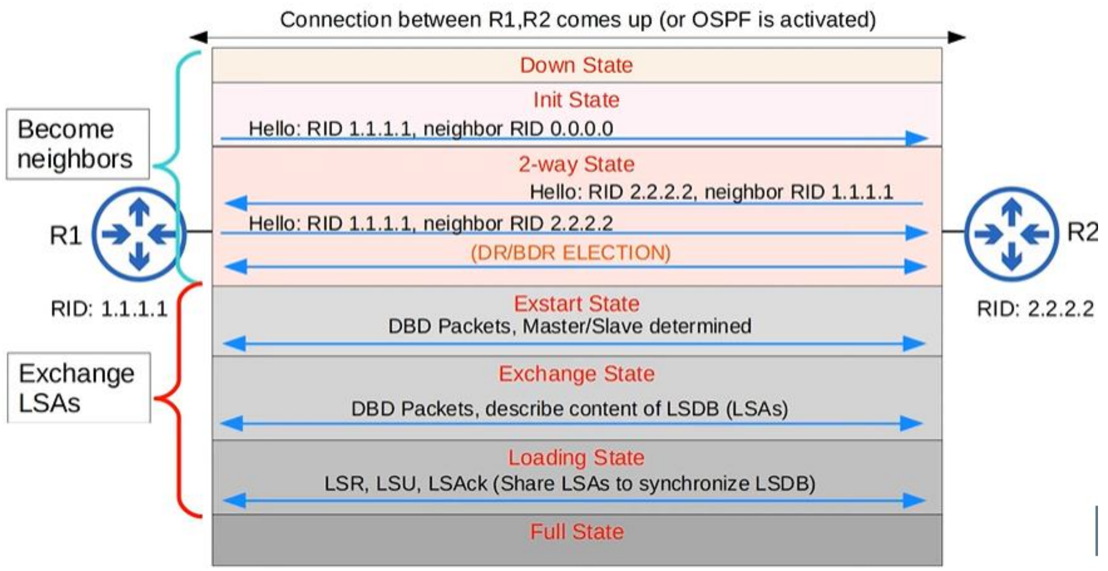
- Send Hello packets every 10 seconds
- Reset the Dead timer on each Hello

Dead timer:

- Default is 40 seconds
- If it reaches 0, the neighbour is removed

Routers continue exchanging LSAs as the network changes to maintain an accurate LSDB.

Summary of Neighbour State Progression



OSPF Message Types Summary

Type	Name	Purpose
1	**Hello**	Neighbor discovery and maintenance.
2	**Database Description (DBD)**	Summary of the LSDB of the router. Used to check if the LSDB of each router is the same.
3	**Link-State Request (LSR)**	Requests specific LSAs from the neighbor.
4	**Link-State Update (LSU)**	Sends specific LSAs to the neighbor.
5	**Link-State Acknowledgement (LSAck)**	Used to acknowledge that the router received a message.

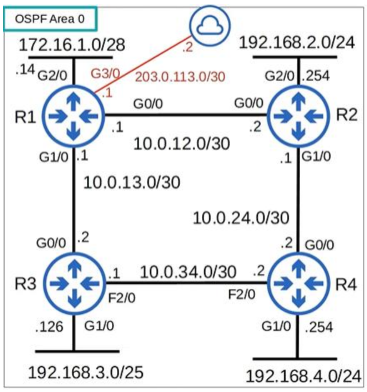
These message types are numbered 1 through 5.

Reviewing OSPF Show Commands

After the overview, let's look again at a few OSPF show commands. You should understand their output better now.

SHOW IP OSPF NEIGHBOR:

This command was entered on R1.




```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/DR         00:00:39    10.0.13.2       GigabitEthernet1/0
2.2.2.2           1   FULL/DR         00:00:31    10.0.12.2       GigabitEthernet0/0
```

Explanation:

- Both neighbours R2 and R3 are in the Full state.
- Both R2 and R3 are Designated Routers (DRs).
- The Dead Time counts down from 40 seconds.

- As soon as a Hello packet is received, the Dead timer resets.
- Assuming Hellos every 10 seconds, the timer:
  - Counts down to ~30
  - Resets to 40
  - Repeats this cycle

SHOW IP OSPF INTERFACE (R1 G0/0):

Now let's look at SHOW IP OSPF INTERFACE on R1's G0/0.

CLI Output:

```
R1#show ip ospf interface g0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.0.12.1/30, Area 0, Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 10.0.12.2
  Backup Designated router (ID) 1.1.1.1, Interface address 10.0.12.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:07
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

Explanation:

- Default Hello and Dead timers are 10 and 40.
- "Hello due in 7 seconds" means:
  - R1 will send a Hello in 7 seconds
  - Hellos are sent every 10 seconds
- Neighbour count is 1
- Adjacent neighbour count is 1
- R1 has only one neighbour on G0/0: R2
- R2 is shown as the Designated Router
- Difference between neighbour and adjacent neighbour will be explained late

More OSPF Configuration

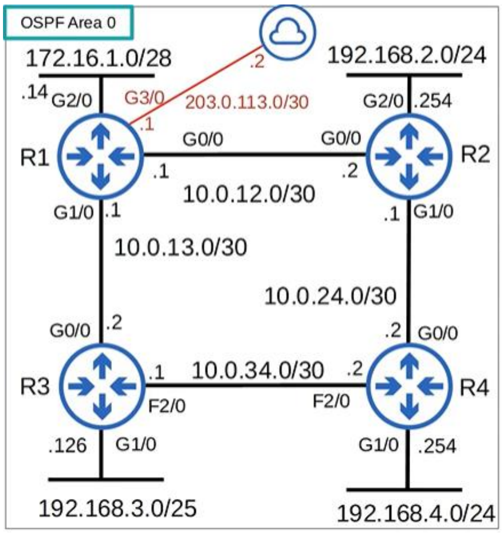
Activating OSPF Directly on an Interface

Let's move on to more OSPF configuration.

Recall the purpose of the NETWORK command:

- Same in RIP, EIGRP, and OSPF
- It tells the router which interfaces to activate the routing protocol on

You can enable OSPF directly on an interface, without using the NETWORK command.



Assume R1 has no OSPF configuration yet.

CLI Configuration:

```
R1(config)#int g0/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int g1/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int g2/0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#int l0
R1(config-if)#ip ospf 1 area 0
```

Explanation:

- Command format:
  - ip ospf process-id area area-id
- This is done in interface configuration mode
- OSPF is enabled on the interfaces
- OSPF configuration mode is not required

Alternative Method to Configure Passive Interfaces

Another method exists to configure passive interfaces.

CLI Configuration:

```
R1(config-if)#router ospf 1
R1(config-router)#passive-interface default
R1(config-router)#no passive-interface g0/0
R1(config-router)#no passive-interface g1/0
```

Explanation:

- passive-interface default makes all interfaces passive
- no passive-interface is used on specific interfaces to make them active
- This is another way to configure passive interfaces
- Depending on the situation:
    - This method may be faster
    - Or the normal method may be faster
- The effect is the same

SHOW IP PROTOCOLS (Interface-Based OSPF)

If OSPF is configured directly on interfaces, the output of SHOW IP PROTOCOLS changes slightly.

CLI Output:

```
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    GigabitEthernet1/0
    GigabitEthernet0/0
    GigabitEthernet2/0
  Passive Interface(s):
    Ethernet0/0
    GigabitEthernet2/0
    GigabitEthernet3/0
    Loopback0
    VoIP-Null0
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110      00:09:53
    Gateway         Distance      Last Update
    3.3.3.3              110      00:09:54
    4.4.4.4              110      00:09:54
  Distance: (default is 110)
```

Explanation

- The **routing for networks** section is empty

```
Routing for Networks:
```

- Instead, interfaces are listed under:
    - “Routing on interfaces configured explicitly”

```
Routing on Interfaces Configured Explicitly (Area 0):
  Loopback0
  GigabitEthernet1/0
  GigabitEthernet0/0
  GigabitEthernet2/0
```

- The rest of the output is the same

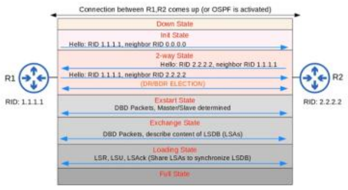
Summary

OSPF Metric (Cost)

- OSPF's metric is cost
- By default:
  - Cost = reference bandwidth ÷ interface bandwidth
- Values less than 1 are converted to 1
- Default reference bandwidth is 100
- Interfaces ≥100 Mbps all have cost 1
- You can change:
  - Reference bandwidth with auto-cost reference-bandwidth
  - Interface cost with ip ospf cost
  - Interface bandwidth with bandwidth (not recommended)
- A route's metric is the sum of outgoing interface costs

OSPF Neighbour Process:

- Routers go through multiple neighbour states
- This is the most difficult part of the lecture
- Recommended to:
  - Watch multiple times
  - Research "OSPF neighbour states"



Additional OSPF Configurations:

- OSPF can be activated directly on interfaces:
  - ip ospf process-id area area-id
- Passive interfaces can be configured with:
  - passive-interface default
  - no passive-interface <interface>